



PHY 320

Modern Physics

Dr. Michelle P. Kuchera | Dana 269 | mikuchera@davidson.edu

MWF 9:40-10:30 | Watson 119 | Lab Mon or Tues

What is Modern Physics?

You've done the basics... now the fun stuff!!

Modern Physics covers scientific advancement from special relativity to today's "hot topics" in physics research. We will discuss recent discoveries, and reproduce key "tabletop" experiments that changed our understanding of the universe we live in.

Topics include relativity, quantum mechanics, nuclear physics, and more! Any of the following fulfills the **prerequisite**: PHY 220/225/230, or instructor permission.

It is important to acknowledge that each of us has a responsibility to create an environment where we can all learn from and teach each other. We are more successful when everyone participates since we all benefit from the unique insights and experiences that each person brings to the class.

Fall 2021 Notice

We are entering yet another semester where there is much uncertainty in the world. Please prioritize your health [physical and mental] and keep notice that your classmates (and professor) are doing the same. Empathy and compassion are essential to establish "lives of leadership and service". We will adapt to the tide of the semester as we progress, so contents of this syllabus may change in response to our experiences.

	Laboratory	You must participate in lab every week. The grade is based off of your lab reports.	30%
	Problem Sets	These are a mixture of group and solo problems related to content covered in class	25%
	In-class Activities	Computational activities, written activities, and class presentations	20%
	Assessments	Each module has a short assessment	25%

Grade Breakdown

Additionally, you will have the option to improve a subset of your module assessments during finals



Most Modern of Physics

This class is about **Modern Physics**. If you notice a topic in the news, on social media, or anywhere else that you would like discussed in class, email Dr. Kuchera with a link to the physics sometime before class, and she will plan to allocate time towards this topic.

tentative schedule

module	
1	Relativity Particle Physics
2	Quantum Mechanics
3	Nuclear Physics Atomic Physics
4	Statistical Physics Solids
4	General Relativity Cosmology

accommodations religion

If you anticipate assignments or other course deadlines conflicting with a major religious holiday of your tradition, please let me know. We will work together to make the necessary accommodations.

access

The college welcomes requests for accommodations related to disability and will grant those that are determined to be reasonable and maintain the integrity of a program or curriculum. To make such a request or to begin a conversation about a possible request, please contact the Office of Academic Access and Disability Resources, which is located in the Center for Teaching and Learning in the E.H. Little Library: Beth Bleil, Director, bebleil@davidson.edu, 704-894-2129; or Alysén Beaty, Assistant Director, albeaty@davidson.edu, 704-894-2939. It is best to submit accommodation requests within the drop/add period; however, requests can be made at any time in the semester. Please keep in mind that accommodations are not retroactive.

Learning Objectives

By the end of this course, you will:

1. obtain a deeper conceptual understanding of modern physical phenomena;
2. develop more sophisticated analytical problem-solving skills, including advanced calculus;
3. conduct laboratory experiments on a variety of quantum mechanical systems and effects;
4. apply modern concepts to interpret new and unusual experimental results;
5. analyze and model the physics with computational resources;

Learning Outcomes

By the end of this course, you will:

1. predict the outcomes of observations that require ideas developed in the modern era;
2. explain the behavior of modern physical systems;
3. solve intermediate-level quantitative problems involving concepts in modern physics;
4. demonstrate experimental familiarity with a variety of modern physical phenomena;
5. configure and operate modern laboratory instruments, with an emphasis on spectroscopy
6. analyze data and compute theoretical results computationally